Based on an analysis of the provided codebase, several refactorings, architectural improvements, and feature additions can enhance maintainability, testability, and scalability.

1. Package and Naming Refactorings

- **Package Name Consistency:** The instrumentation test file path uses `com.example.pdrwithekftest`, while the rest of the app and unit tests use `com.example.pdrandroutetest`. These should be unified to prevent confusion or classpath mismatches. TXT + 1
- **Class Renaming (`DiagnosticEkf`):** Since this class acts as the core EKF implementation used in production code as well as diagnostics, renaming it to `PdrEkfEngine` or `SensorFusionEkf` would reflect its active runtime role rather than just diagnostic testing. TXT
- **File Separation (`MapScreen.kt`):** The `ViewportTracker` class is defined directly inside `MapScreen.kt`. Moving `ViewportTracker` into its own file under a `ui/viewport/` or `util/` package will improve separation of concerns. TXT + 1

2. Architectural Improvements and Code Smells

- **Decouple the "God Object" (`HikeRepository`):** `HikeRepository` currently acts as a massive singleton handling EKF state, route management, grid display preferences, UI timers, TTS messages, navigation logs, bearing corrections, and controlling a playback engine. TXT
  - *Improvement:* Split these responsibilities into distinct modules or repositories, such as a `NavigationStateRepository`, `MapPreferencesRepository`, and `HikeTracker`.
- **Simplify Concurrency Models:** `TelemetryPlaybackEngine` mixes Java's `@Synchronized` annotations, `volatile` fields, and Kotlin Coroutines `Mutex` locks. Standardizing entirely on Kotlin coroutines (`Mutex` or structured concurrency) will make state synchronization easier to reason about and less error-prone. TXT + 1
- **Extract Hardcoded Thresholds:** Magic numbers like the 15-meter arrival threshold, 100-meter approach threshold, and 10-degree off-course deviation threshold are hardcoded across `Hike.kt` and `Waypoint.kt`. Moving these into a centralized `HikeConfig` data class would make them configurable. TXT + 1

3. Function Parameter List Optimizations

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- **Refactor `ControlPanel` Parameter List**: The `ControlPanel` Composable takes 17 individual parameters. TXT
  - *Improvement:* Group these parameters into dedicated state data classes, such as `ControlPanelState` and `ControlPanelCallbacks`, to clean up the function signature and improve readability. TXT
- **Refactor `Hike.evaluateLegState` and `stringify`**: These methods accept multiple optional parameters (`ekfState`, `currentTimeMs`, `location`). Grouping evaluation context into a `HikeEvaluationContext` data class would simplify future extensions. TXT + 1

4. Code Cleanups

- **Remove Redundant Fallback Checks**: In `Hike.kt`, `HIKE_ENDED` is checked dynamically inside `evaluateLegState` through multiple conditional branches that overlap with target distance checks. Streamlining this logic will reduce redundant conditional branching. TXT + 1
- **Consolidate Coordinate Conversion Logic**: Coordinate utility functions are split between `CoordinateUtils` and `IrishGridConverter`. Encapsulating private internal helpers more cleanly will make the conversion boundary easier to unit test. TXT + 1

5. Suggested New Functionality

- **GPX / KML Route Export/Import**: Currently, routes only support custom CSV files. Adding support for standard GPX or KML file formats would make it easier to import real-world hiking trails from third-party apps. TXT + 1
- **Compass Calibration Status Alerts**: The app tracks compass bias (`b_theta`) in the EKF. Exposing a visual warning in the `TelemetryDashboard` when compass variance exceeds safe thresholds would alert users to magnetic interference. TXT + 1
- **Hike Summary Report Screen**: When a hike finishes or playback completes, generating a structured summary screen showing total distance, average stride length, step counts, and GPS correction frequencies would provide a polished user experience.